

Upward revision of global fossil fuel methane emissions based on isotope database

Stefan Schwietzke^{1,2}, Owen A. Sherwood³, Lori M. P. Bruhwiler², John B. Miller^{1,2}, Giuseppe Etiope^{4,5}, Edward J. Dlugokencky², Sylvia Englund Michel³, Victoria A. Arling^{1,2}, Bruce H. Vaughn³, James W. C. White³ & Pieter P. Tans²

Methane has the second-largest global radiative forcing impact of anthropogenic greenhouse gases after carbon dioxide, but our understanding of the global atmospheric methane budget is incomplete. The global fossil fuel industry (production and usage of natural gas, oil and coal) is thought to contribute 15 to 22 per cent of methane emissions^{1–10} to the total atmospheric methane budget¹¹. However, questions remain regarding methane emission trends as a result of fossil fuel industrial activity and the contribution to total methane emissions of sources from the fossil fuel industry and from natural geological seepage^{12,13}, which are often co-located. Here we re-evaluate the global methane budget and the contribution of the fossil fuel industry to methane emissions based on long-term global methane and methane carbon isotope records. We compile the largest isotopic methane source signature database so far, including fossil fuel, microbial and biomass-burning methane emission sources. We find that total fossil fuel methane emissions (fossil fuel industry plus natural geological seepage) are not increasing over time, but are 60 to 110 per cent greater than current estimates^{1–10} owing to large revisions in isotope source signatures. We show that this is consistent with the observed global latitudinal methane gradient. After accounting for natural geological methane seepage^{12,13}, we find that methane emissions from natural gas, oil and coal production and their usage are 20 to 60 per cent greater than inventories^{1,2}. Our findings imply a greater potential for the fossil fuel industry to mitigate anthropogenic climate forcing, but we also find that methane emissions from natural gas as a fraction of production have declined from approximately 8 per cent to approximately 2 per cent over the past three decades.

Our current understanding of the global methane (CH₄) budget stems largely from three-dimensional (3D) inversion studies, which use the trends and gradients of atmospheric CH₄ to infer the spatio-temporal distribution of the total CH₄ source, but atmospheric CH₄ data alone do not include source type information. Source type information comes primarily from bottom-up-derived a priori spatial patterns. 3D inverse models return a posteriori fluxes constrained by the bottom-up source type information for each, especially for large regions that contain a mix of source types, like agriculture, wetlands and fossil fuels. Note that we refer below to CH₄ emissions from total fossil fuels (FF_{tot}) as the sum of CH₄ emissions from fossil fuel industry activities (FF_{ind}) and natural geological seepage (FF_{geo}). To alleviate this problem, some previous 3D inversion^{3,4} and box model studies^{9,10} have included measurements of atmospheric $\delta^{13}\text{C}-\text{CH}_4$ (henceforth $\delta^{13}\text{C}_{\text{atm}}$, where $\delta^{13}\text{C} = R_{\text{atm}}/R_{\text{std}} - 1$ and $R = {}^{13}\text{C}/{}^{12}\text{C}$) as an additional constraint for better source allocation. Broadly defined source categories—that is, FF_{tot}, microbial, and biomass burning—emit CH₄ with different source signatures¹⁰ ($\delta^{13}\text{C}-\text{CH}_4$; henceforth $\delta^{13}\text{C}_{\text{source}}$, including $\delta^{13}\text{C}_{\text{FF}}$, $\delta^{13}\text{C}_{\text{mic}}$ and $\delta^{13}\text{C}_{\text{BB}}$).

The sample sizes of $\delta^{13}\text{C}_{\text{source}}$ values used in published global CH₄ budgets are either small ($N < 100$, based on cited original measurements) or unknown, uncertainties are rarely applied, and

global representativeness is lacking (especially in the tropics and the Southern Hemisphere), but some $\delta^{13}\text{C}_{\text{source}}$ values have nevertheless taken on canonical status with few references to primary sources (for example, refs 3, 4, 9 and 10; see full list of references in Supplementary Information section 8). We have compiled the most comprehensive $\delta^{13}\text{C}_{\text{source}}$ database to date (see ref. 14 and Supplementary Information sections 3–5 for complete list of data, metadata and references) including 9,468 $\delta^{13}\text{C}_{\text{FF}}$, $\delta^{13}\text{C}_{\text{mic}}$ and $\delta^{13}\text{C}_{\text{BB}}$ original measurements from the peer-reviewed literature and other publicly available sources to define globally weighted average $\delta^{13}\text{C}_{\text{FF}}$ (time-dependent), $\delta^{13}\text{C}_{\text{mic}}$, and $\delta^{13}\text{C}_{\text{BB}}$ with well defined uncertainties. These data allowed us to revisit the source attribution of global CH₄ emissions since the 1980s by applying an atmospheric box-model to global atmospheric CH₄ and $\delta^{13}\text{C}_{\text{atm}}$ measurements (and avoiding the use of a priori FF_{tot} and microbial source strengths), thus maximizing the CH₄ and $\delta^{13}\text{C}_{\text{atm}}$ constraints.

Our box-model applies Monte Carlo techniques to estimate global FF_{tot} and microbial CH₄ emissions and uncertainties as a function of $\delta^{13}\text{C}_{\text{source}}$, of isotope fractionation during oxidation (OH + CH₄), of the uncertainties of both of these values, and of other factors (see Supplementary Table 1). We also estimated FF_{ind} emissions by subtracting FF_{geo} emissions from FF_{tot} emissions. This allowed us to calculate global long-term trends in the Fugitive Emission Rate (FER), which is the fraction of natural gas production lost to the atmosphere through its lifecycle (production, processing, transport and use), and is a critical parameter for evaluating the climatic impact of natural gas as a fuel^{9,15}.

The $\delta^{13}\text{C}_{\text{source}}$ weighting procedures are described in detail in Supplementary Information sections 3, 4 and 5, and briefly summarized here. The $\delta^{13}\text{C}_{\text{FF}}$ samples ($N = 7,482$) are representative of natural gas or coal gas from 44 countries, accounting for 82% and 80% of global natural gas and coal production, respectively¹⁶. Country-specific $\delta^{13}\text{C}_{\text{FF}}$ distributions for natural gas and coal were weighted by their respective annual production of natural gas, oil (co-produced with natural gas), and coal (Supplementary Information section 5). The time averaged, globally weighted $\delta^{13}\text{C}_{\text{FF}}$ of $-44.0 \pm 0.7\text{‰}$ (one standard deviation, 1 s.d.) is much lighter (about 5‰ lighter) than typical values in previous inverse studies (Fig. 1), although Whiticar *et al.*¹⁷ reported an empirically derived $\delta^{13}\text{C}_{\text{natural gas/oil}}$ value of -44‰ based on unpublished proprietary industry sources. Our relatively light $\delta^{13}\text{C}_{\text{natural gas/oil}}$ value is due to contributions from economically important reservoirs of microbially produced CH₄, and from thermogenic gas originating from low-maturity source rocks, or is associated with oil, typically¹⁸ in the range -45‰ to -55‰ . Thermogenic CH₄ formation and microbial methanogenesis also occur in coal beds (both deep and shallow deposits¹⁹).

Our $\delta^{13}\text{C}_{\text{source}}$ database¹⁴ contains 1,021 $\delta^{13}\text{C}_{\text{mic}}$ samples from wetlands, termites, ruminants, rice agriculture and waste/landfill, weighted by their relative contribution to global microbial CH₄

¹Cooperative Institute for Research in Environmental Sciences, University of Colorado, Boulder, Colorado, USA. ²NOAA Earth System Research Laboratory, Global Monitoring Division, Boulder, Colorado, USA. ³Institute of Arctic and Alpine Research, University of Colorado, Boulder, Colorado, USA. ⁴Istituto Nazionale di Geofisica e Vulcanologia, Sezione Roma 2, Italy. ⁵Faculty of Environmental Science and Engineering, Babes Bolyai University, Cluj-Napoca, Romania.

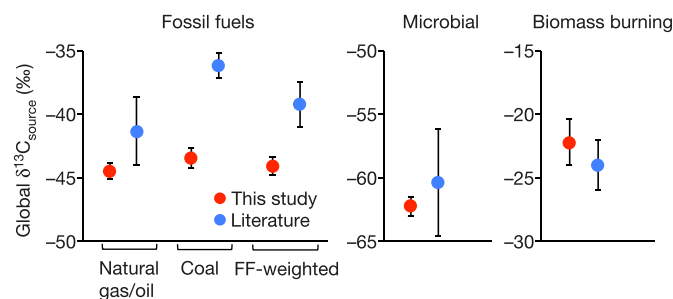


Figure 1 | Comparison of $\delta^{13}\text{C}_{\text{FF}}$, $\delta^{13}\text{C}_{\text{mic}}$ and $\delta^{13}\text{C}_{\text{BB}}$ source signatures from this study (red) and those used in 15 previous top-down studies (blue; see Supplementary Information section 8 for references). Error bars indicate 1 s.d., representing empirical uncertainty in our database (red; $N_{\text{total}} = 9,468$; see Supplementary Table 1) and variability among mean literature values (blue). $\delta^{13}\text{C}_{\text{FF}}$ are temporal averages; see Supplementary Fig. 11 for annual $\delta^{13}\text{C}_{\text{FF}}$ (owing to country-specific FF_{ind} production trends).

emissions based on 12 literature estimates (Supplementary Information section 3). Our $\delta^{13}\text{C}_{\text{source}}$ database¹⁴ includes 82 direct (plume measurement) and 965 indirect (plant material) $\delta^{13}\text{C}_{\text{BB}}$ samples from C_3 and C_4 plants, weighted by maps of global vegetation and biomass-burning fluxes (Supplementary Information section 4). Our $\delta^{13}\text{C}_{\text{mic}}$ and $\delta^{13}\text{C}_{\text{BB}}$ are within the uncertainty of literature estimates, but have smaller error bars (given the large sample size), and approximately 2‰ differences in mean values (Fig. 1).

Time series of the global microbial and FF_{tot} CH_4 emission distributions are shown in Fig. 2a using 10,000 Monte Carlo box-model

repetitions. Estimates of FF_{tot} CH_4 are based only on global average atmospheric CH_4 levels, CH_4 lifetimes, isotope source signatures, isotope fractionation, and soil sink and biomass-burning estimates; that is, they exclude other bottom-up estimates or a priori information. Note that this study focuses on long-term trends, and Fig. 2a presents moving averages. The original model results (Supplementary Fig. 18) include the inter-annual variability of <10% on average, which is partly an artefact of multiple components that our model does not control including CH_4 sink inter-annual variability and the $\delta^{13}\text{C}_{\text{BB}}$ inter-annual variability depending on the dominant biomass type (C_3 versus C_4) in a given year.

Our updated $\delta^{13}\text{C}_{\text{source}}$ database causes a substantial upward shift in total fossil fuel CH_4 relative to 'traditional' $\delta^{13}\text{C}_{\text{source}}$ used in the literature. The approximately 5‰ lighter $\delta^{13}\text{C}_{\text{FF}}$ alone leads to a FF_{tot} value about 50 $\text{Tg CH}_4 \text{ yr}^{-1}$ greater, and the combination of the approximately 2‰ lighter $\delta^{13}\text{C}_{\text{mic}}$ and approximately 2‰ heavier $\delta^{13}\text{C}_{\text{BB}}$ shifts FF_{tot} up by an additional 25 $\text{Tg CH}_4 \text{ yr}^{-1}$ or so (Fig. 2a). Total annual CH_4 emissions increased by about 25 Tg since 2006 (ref. 20, Supplementary Fig. 2), and the microbial source increased by about 45 Tg , partially offset by a FF_{tot} decrease of about 20 Tg to balance the atmospheric CH_4 budget. This growth attribution to microbial sources is mostly driven by changing $\delta^{13}\text{C}_{\text{atm}}$ (increase stopped around 2000 followed by a 0.2‰ decrease since 2004; see also ref. 21), and a 1.7‰ $\delta^{13}\text{C}_{\text{FF}}$ increase since 2003 resulting from a redistribution of global FF_{ind} production from countries with different $\delta^{13}\text{C}_{\text{FF}}$ values. For example, China's rising hard coal production during 1999–2012 increased global coal $\delta^{13}\text{C}_{\text{FF}}$ by approximately 3.5‰ given China's national average coal $\delta^{13}\text{C}_{\text{FF}}$ of -36.0‰ , and global natural gas $\delta^{13}\text{C}_{\text{FF}}$ increased by approximately 0.7‰ during 2002–2012 because of rising natural gas

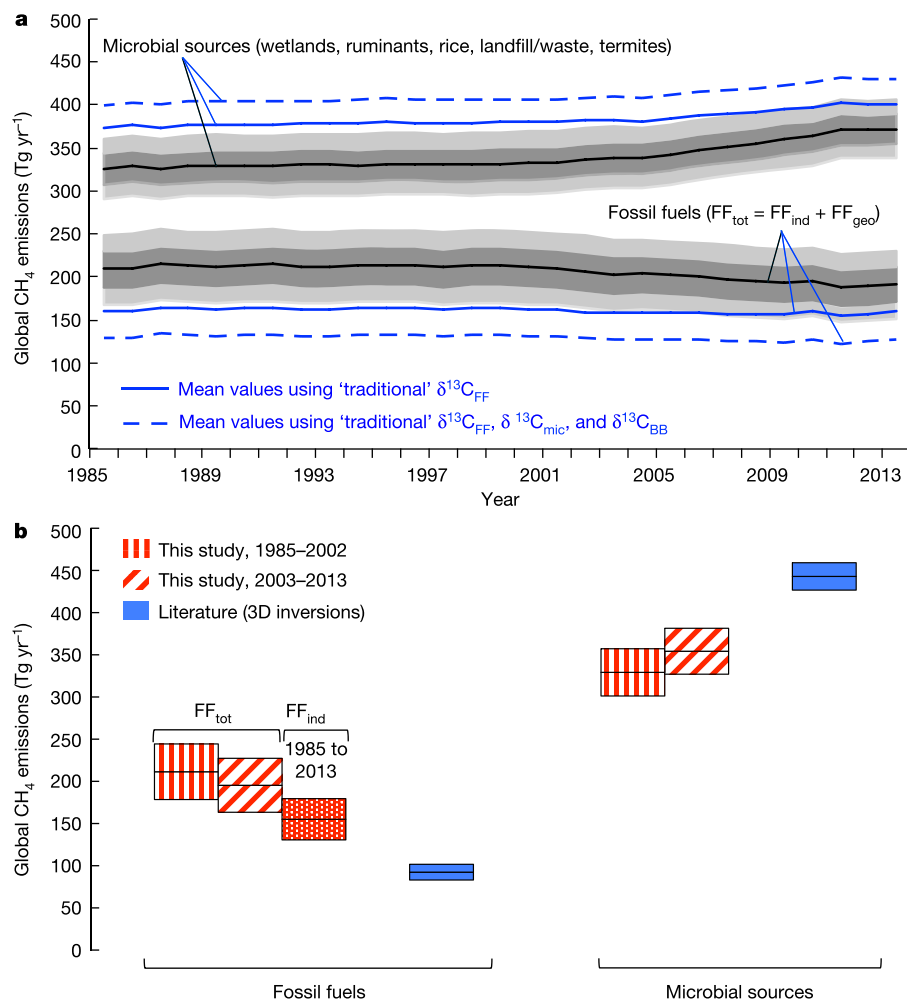


Figure 2 | Fossil fuel and microbial source CH_4 budget terms. **a**, Long-term trend in global microbial and total FF_{tot} CH_4 emissions from 1985 to 2013. Moving averages are shown; see Supplementary Fig. 18 for original mass balance results including inter-annual variability due to multiple components not accounted for in this model (see text). Mean values are shown in solid black. Dark and light grey bands mark the 25th/75th percentile and the 10th/90th percentiles, respectively. Blue lines assume the mean $\delta^{13}\text{C}_{\text{source}}$ values from the literature (as in Fig. 1, blue values). See Supplementary Information section 7 for sensitivity analyses. **b**, The box plot compares means and 1 s.d. uncertainties between this study (red) and the recent literature^{3–8} 3D inversions (blue). The box-model temporal split marks approximately when the $\delta^{13}\text{C}_{\text{atm}}$ increase stopped and $\delta^{13}\text{C}_{\text{FF}}$ decreased; literature periods vary. The literature budget terms were scaled to match this study's mean total CH_4 budget (see Supplementary Information section 8 for individual literature study means). Industrial FF (FF_{ind}) represents total FF (FF_{tot}) minus natural geological seepage (FF_{geo}). Literature $\text{FF}_{\text{tot}} = \text{FF}_{\text{ind}}$ because these studies exclude FF_{geo} .

production in multiple countries with relatively heavy natural gas $\delta^{13}\text{C}_{\text{FF}}$ (details in Supplementary Information section 3).

The FF_{tot} CH_4 decrease is surprising because it marks a period of a dramatic (56%) increase in global coal production, mainly from China¹⁶. However, not accounting for other factors in our model could produce a downward bias of up to 20 Tg in the FF_{tot} CH_4 trend (Supplementary Fig. 17 for a sensitivity analysis of the model parameters on the budget terms and their trends). For instance, the 0.2‰ $\delta^{13}\text{C}_{\text{atm}}$ decrease requires decreasing emissions of either FF_{tot} or biomass burning since 2004, and we used a prescribed, temporally constant range of biomass-burning emissions. Constant or decreasing FF_{tot} emissions disagree with emission inventories, suggesting increasing FF_{ind} emissions during this period^{1,21}. This may be explained by efficiency improvements such as capturing otherwise vented or flared natural gas, replacement of old equipment, and improved combustion techniques²², which inventories may not account for completely.

Box-model and literature 3D inverse estimates^{3–8} are compared in Fig. 2b. Mean FF_{tot} CH_4 emissions in this study are approximately double previous estimates. The relatively small size of the 3D inverse FF_{tot} CH_4 estimates is partly due to the strong influence of a priori information on source attribution in regions where observations are sparse and multiple sources are co-located. As a result, the inverse FF a posteriori means are on average within 17% of the a priori means^{3–8}. Also, these studies are not independent owing to their choice of broadly similar a priori information (these studies use different releases of the EDGAR database² for anthropogenic emissions): the a priori mean of five^{3–6,8} out of the six studies (Fig. 2b) is only 92 ± 3 Tg CH_4 yr^{-1} (1 s.d.). Note that when $\delta^{13}\text{C}_{\text{atm}}$ is used in 3D inversions^{3,4} (as opposed to box-model studies), the spatio-temporal CH_4 constraint greatly outweighs the information in the relatively sparse $\delta^{13}\text{C}_{\text{atm}}$ data. For instance, the relative difference in a posteriori FF_{tot} fluxes between including and excluding $\delta^{13}\text{C}_{\text{atm}}$ data in 3D inversions^{3,4} is <7%. Uncertainties in this study (the FF_{tot} 1 s.d. is 32 Tg CH_4 yr^{-1}) are considerably larger than in the inverse studies (9 Tg CH_4 yr^{-1} on average). However, inversion-based a posteriori uncertainties are often derived from relatively narrow a priori source uncertainties, whereas we derive uncertainties directly from sensitivities in the global mass balances of $\delta^{13}\text{C}_{\text{atm}}$ and CH_4 . The only prescribed source in our model is biomass burning, although our prescribed biomass-burning estimates are partly constrained by fire detection using satellites (wild fires can be detected²³, not accounting for fuel biomass burning). These results suggest the route of first obtaining observation-based global total CH_4 source strengths (this study) as inputs to an inversion, which can then use additional spatial information to estimate source allocation at higher resolution.

Until now, most top-down studies have excluded FF_{geo} as an important source of global CH_4 over the past three decades¹¹, despite bottom-up studies^{12,13} as well as top-down analyses using ice-core^{24–26} and radiocarbon (^{14}C) data¹² suggesting a FF_{geo} range of 20–76 Tg CH_4 yr^{-1} (4%–14% of the modern global budget; Supplementary Information sections 2 and 6). Constraining these emissions using ice-core CH_4 and $\delta^{13}\text{C}$ (henceforth $\delta^{13}\text{C}_{\text{ice}}$) measurements, and our extensive database of isotope signatures yields FF_{geo} emissions of 51 ± 20 Tg CH_4 yr^{-1} (1 s.d.; Supplementary Information section 6). Subtracting FF_{geo} from FF_{tot} yields mean FF_{ind} emissions of 156 ± 24 Tg CH_4 yr^{-1} (the 1 s.d. uncertainty accounts for the correlation coefficient of 1 between FF_{tot} and FF_{geo} as described in Supplementary Information section 1), that is, still 65% greater than previous 3D inverse model estimates.

By mass balance, the microbial source (23% smaller than the literature^{3–8}) must account for the difference between global CH_4 emissions, FF_{tot} , and biomass burning. Thus, our results suggest an important shift in the current understanding of the global CH_4 budget towards a higher FF_{tot} component compensated by lower microbial emissions, but the recent temporal increases in microbial emissions

Table 1 | $\delta^{13}\text{C}$ -based source attribution means for different periods.

	0–1700 AD*	1985–2002 AD	2003–2013 AD
Total fossil fuels†	51 ± 20	211 ± 33	195 ± 32
Fossil fuel industries	0	161 ± 24	145 ± 23
Geological sources		51 ± 20	
Microbial	154 ± 19	330 ± 28	355 ± 27
Biomass burning	25 ± 5		43 ± 9

Values are given as mean $\pm$ one standard deviation in units of teragrams of methane per year.

The biomass burning ranges are those prescribed in the $\delta^{13}\text{C}$ mass balance.

*See text and Supplementary Information section 6.

†TM5 simulations of the latitudinal gradient and comparison with observations indicate a present-day FF_{tot} range of 150–200 Tg CH_4 yr^{-1} (see text and Supplementary section 7).

have been substantially larger. The $\delta^{13}\text{C}$ mass balance approach cannot distinguish between natural and anthropogenic microbial sources. However, the magnitude of the anthropogenic microbial sources (agriculture, including livestock and rice production, and waste, including landfill) has historically been related to population growth, and Schaefer *et al.*²¹ recently found it plausible that agriculture and waste account for some of the temporal CH_4 emissions increase based on $\delta^{13}\text{C}$. All estimated pre-industrial and present-day global CH_4 budget terms are summarized in Table 1.

We further evaluated our $\delta^{13}\text{C}$ -based source attribution by simulating global atmospheric CH_4 mole fractions using emission maps scaled by the $\delta^{13}\text{C}$ -based source attribution, and transported by the 3D global atmospheric chemistry and transport model TM5²⁷. The simulated and observed global north–south gradients of CH_4 at remote background sites of NOAA's Global Greenhouse Gas Reference Network²⁰ add a spatial source attribution constraint because the broad spatial distribution of some CH_4 source categories is relatively well known globally. On the basis of nine simulated scenarios, we find that FF_{tot} in the range 150–200 Tg CH_4 yr^{-1} is consistent with the observed global north–south gradient (Supplementary Information section 7).

Inferred natural-gas industry efficiency improvements are illustrated using the FER time series in Fig. 3, which is calculated as FF_{ind} emissions (FF_{tot} minus FF_{geo}) minus oil and coal emissions (bottom-up estimates including uncertainties²⁸) divided by global dry gas production of natural gas¹⁶ and accounting for the CH_4 content of natural gas (Supplementary Information section 1). Mean FER decreases from 7.6% in 1985 to 2.2% in 2013. Assuming 'traditional' $\delta^{13}\text{C}_{\text{source}}$ values for all sources (blue dashed lines) would yield negative mean FER in some years, which is impossible, thereby emphasizing the importance of the updated $\delta^{13}\text{C}_{\text{source}}$. Note that the oil and coal emissions used to estimate FER assume temporally constant oil and coal emission factors while production increased, that is, no efficiency

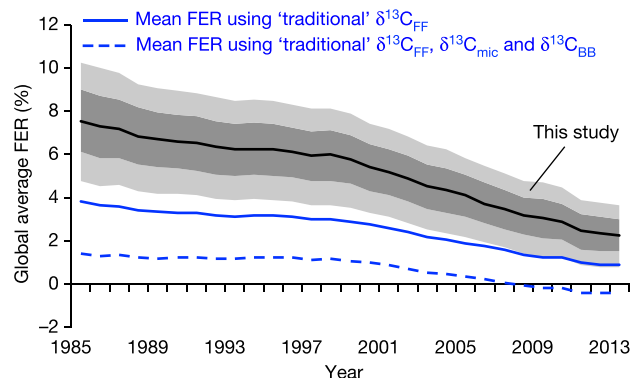


Figure 3 | Global FER long-term trend with mean values shown in solid black. Moving averages are shown; see Supplementary Fig. 18 for original mass balance results including inter-annual variability due to multiple components not accounted for in this model (see text). Dark and light grey bands mark the 25th/75th percentile and the 10th/90th percentile uncertainties, respectively. The dashed black line represents the linear trend of the means. Blue lines assume the mean $\delta^{13}\text{C}_{\text{source}}$ values from the literature (as in Fig. 1, blue values).

improvements in the oil and coal industries. Thus, the FER decline rate would be smaller in the case of noticeable oil and coal efficiency improvements. However, the impact of potential oil CH₄ emission reductions would be minor because oil contributes only on average 10% to FF_{ind} CH₄ emissions (Supplementary Fig. 10). Global coal CH₄ mitigation from reported projects, which may not be included in the bottom-up estimates amounts to only around 7% of global coal CH₄ emissions^{1,29}.

Our finding that FF_{tot} CH₄ emissions are 60%–110% higher than previous studies based on the most comprehensive global $\delta^{13}\text{C}_{\text{source}}$ database compiled so far represents a major adjustment in the global CH₄ budget, and this is consistent with the observed latitudinal CH₄ gradient. It agrees with a previous radiocarbon analysis³⁰ ($167 \pm 13 \text{ Tg CH}_4 \text{ yr}^{-1} \text{ FF}_{\text{tot}}$), which had so far been considered a “plausible re-estimate rather than a definitive revision”³⁰ of FF_{tot} owing to the model complexity involved. Our revised FF_{tot} emissions are compensated by lower microbial CH₄ emissions, and this is consistent with the palaeo-CH₄ budget (Supplementary Information section 6).

Accounting for previously neglected FF_{geo}, our correction of 20%–60% higher CH₄ emissions from natural gas, oil and coal production and use implies a greater potential for industry efficiency improvements to mitigate anthropogenic climate forcing. Yet, this study does not confirm an upward trend of FF_{ind} emissions in global CH₄ inventories^{1,2} despite the large increase in natural gas, oil and coal production and use over the past three decades. Instead, this study finds that natural-gas CH₄ emissions per unit of production have declined from about 8% in the mid-1980s to about 2% in the late 2000s and early 2010s. Natural-gas industry improvements associated with management practices, technology, and replacement of older equipment have been credited with reducing CH₄ leakage in the past¹. The global observations used in our study confirm this trend, but the industry improvements have been offset by increased natural-gas production. Ongoing and future field studies at the level of natural-gas basins, facilities and components may help us to understand the contribution of individual leak types in order to reduce total natural-gas CH₄ emissions.

Received 25 April; accepted 22 August 2016.

- US Environmental Protection Agency. *Summary Report: Global Anthropogenic Non-CO₂ Greenhouse Gas Emissions: 1990–2030* https://www3.epa.gov/climatechange/Downloads/EPAactivities/Summary_Global_NonCO2_Projections_Dec2012.pdf (Office of Atmospheric Programs, Climate Change Division, US EPA, 2012).
- European Commission—Joint Research Centre (JRC)/Netherlands Environmental Assessment Agency (PBL). *Emission Database for Global Atmospheric Research (EDGAR)*, release version 4.2, <http://edgar.jrc.ec.europa.eu> (2014).
- Bousquet, P. *et al.* Contribution of anthropogenic and natural sources to atmospheric methane variability. *Nature* **443**, 439–443 (2006).
- Mikaloff Fletcher, S. E., Tans, P. P., Bruhwiler, L. M., Miller, J. B. & Heimann, M. CH₄ sources estimated from atmospheric observations of CH₄ and its $^{13}\text{C}/^{12}\text{C}$ isotopic ratios: 1. Inverse modeling of source processes. *Glob. Biogeochem. Cycles* **18**, GB4004 (2004).
- Fraser, A. *et al.* Estimating regional methane surface fluxes: the relative importance of surface and GOSAT mole fraction measurements. *Atmos. Chem. Phys.* **13**, 5697–5713 (2013).
- Wang, J. S. *et al.* A 3-D model analysis of the slowdown and interannual variability in the methane growth rate from 1988 to 1997. *Glob. Biogeochem. Cycles* **18**, GB3011 (2004).
- Chen, Y.-H. & Prinn, R. G. Estimation of atmospheric methane emissions between 1996 and 2001 using a three-dimensional global chemical transport model. *J. Geophys. Res. Atmos.* **111**, D10307 (2006).
- Bergamaschi, P. *et al.* Inverse modeling of global and regional CH₄ emissions using SCIAMACHY satellite retrievals. *J. Geophys. Res.* **114**, D22301 (2009).
- Schwietzke, S., Griffin, W. M., Matthews, H. S. & Bruhwiler, L. M. P. Natural gas fugitive emissions rates constrained by global atmospheric methane and ethane. *Environ. Sci. Technol.* **48**, 7714–7722 (2014).
- Quay, P. *et al.* The isotopic composition of atmospheric methane. *Glob. Biogeochem. Cycles* **13**, 445–461 (1999).
- Kirschke, S. *et al.* Three decades of global methane sources and sinks. *Nat. Geosci.* **6**, 813–823 (2013).
- Etiopio, G., Lassey, K. R., Klusman, R. W. & Boschi, E. Reappraisal of the fossil methane budget and related emission from geologic sources. *Geophys. Res. Lett.* **35**, L09307 (2008).
- Ciais, P. *et al.* in *Working Group I Contribution To The IPCC Fifth Assessment Report. Climate Change 2013—The Physical Science Basis* (eds Stocker, T. F. *et al.*) Ch. 6, 465–570 (Cambridge Univ. Press, 2013).
- Sherwood, O., Schwietzke, S., Arling, V. & Etiopio, G. *Global Inventory of Fossil and Non-fossil Methane $\delta^{13}\text{C}$ Source Signature Measurements for Improved Atmospheric Modeling* <http://www.esrl.noaa.gov/gmd/ccgg/d13C-src-inv/> (NOAA/ESRL/GMD, 2016).
- Alvarez, R. A., Pacala, S. W., Winebrake, J. J., Chameides, W. L. & Hamburg, S. P. Greater focus needed on methane leakage from natural gas infrastructure. *Proc. Natl Acad. Sci. USA* **109**, 6435–6440 (2012).
- US Energy Information Administration *International Energy Statistics* database <http://www.eia.gov/cfapps/ipdbproject/IEDIndex3.cfm?tid=3&pid=3&aid=1> (2016).
- Whiticar, M. J. in *Organic Geochemistry* (eds Durand, B. & Behar, F.) Vol. 16, 531–548 (Pergamon, 1990).
- Milkov, A. V. Worldwide distribution and significance of secondary microbial methane formed during petroleum biodegradation in conventional reservoirs. *Org. Geochem.* **42**, 184–207 (2011).
- Meslé, M., Dromart, G. & Oger, P. Microbial methanogenesis in subsurface oil and coal. *Res. Microbiol.* **164**, 959–972 (2013).
- NOAA Earth System Research Laboratory Global Monitoring Division *Global Greenhouse Gas Reference Network* <http://www.esrl.noaa.gov/gmd/ccgg/iadw> (2016).
- Schaefer, H. *et al.* A 21st century shift from fossil-fuel to biogenic methane emissions indicated by $^{13}\text{CH}_4$. *Science* **352**, 80–84 (2016).
- US Environmental Protection Agency. *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2014* <https://www.epa.gov/ghgemissions/us-greenhouse-gas-inventory-report-1990-2014> (2016).
- Randerson, J. T., Chen, Y., van der Werf, G. R., Rogers, B. M. & Morton, D. C. Global burned area and biomass burning emissions from small fires. *J. Geophys. Res.* **117**, G04012 (2012).
- Sapart, C. J. *et al.* Natural and anthropogenic variations in methane sources during the past two millennia. *Nature* **490**, 85–88 (2012).
- Mischler, J. A. *et al.* Carbon and hydrogen isotopic composition of methane over the last 1000 years. *Glob. Biogeochem. Cycles* **23**, GB4024 (2009).
- Sowers, T. Atmospheric methane isotope records covering the Holocene period. *Quat. Sci. Rev.* **29**, 213–221 (2010).
- Krol, M. *et al.* The two-way nested global chemistry-transport zoom model TM5: algorithm and applications. *Atmos. Chem. Phys.* **5**, 417–432 (2005).
- Schwietzke, S., Griffin, W. M., Matthews, H. S. & Bruhwiler, L. M. P. Global bottom-up fossil fuel fugitive methane and ethane emissions inventory for atmospheric modeling. *ACS Sustain. Chem. Eng.* **2**, 1992–2001 (2014).
- International Energy Agency *Coal Mine Methane in China: A Budding Asset with the Potential to Bloom* https://www.iea.org/publications/freepublications/publication/china_cmm_report.pdf (2009).
- Lassey, K. R., Lowe, D. C. & Smith, A. M. The atmospheric cycling of radiomethane and the ‘fossil fraction’ of the methane source. *Atmos. Chem. Phys.* **7**, 2141–2149 (2007).

Supplementary Information is available in the online version of the paper.

Acknowledgements We thank J. Randerson, K. Johnson, A. Cole, C. Iltle, T. Wirth, T. Capehart, S. Montzka, and R. Klusman for comments and discussions. We acknowledge M. Schoell and A. Ionescu for contributing $\delta^{13}\text{C}_{\text{source}}$ data. This research was supported by a National Research Council RAP fellowship and a CIRES IRP grant.

Author Contributions S.S. was responsible for study design, box-model development, analysis of TM5 results, manuscript preparation, and helped provide data. P.P.T. and J.B.M. helped with study design and improved the manuscript. O.A.S., G.E., E.J.D., S.E.M., V.A.A., B.H.V. and J.W.C.W. provided data and improved the manuscript. L.M.P.B. was responsible for TM5 modelling, helped with model analysis, and improved the manuscript.

Author Information The reconstructed atmospheric global average $\delta^{13}\text{C}_{\text{atm}}$ data for 1984–1998 are from ref. 21 and global $\delta^{13}\text{C}_{\text{ice}}$ data are from ref. 24. Reprints and permissions information is available at www.nature.com/reprints. The authors declare no competing financial interests. Readers are welcome to comment on the online version of the paper. Correspondence and requests for materials should be addressed to S.S. (stefan.schwietzke@noaa.gov).

Reviewer Information Nature thanks G. Allen, M. Heimann and the other anonymous reviewer(s) for their contribution to the peer review of this work.

CORRIGENDUM

doi:10.1038/nature21422

Corrigendum: Upward revision of global fossil fuel methane emissions based on isotope database

Stefan Schwietzke, Owen A. Sherwood, Lori M. P. Bruhwiler, John B. Miller, Giuseppe Etiope, Edward J. Dlugokencky, Sylvia Englund Michel, Victoria A. Arling, Bruce H. Vaughn, James W. C. White & Pieter P. Tans

Nature **538**, 88–91 (2016); doi:10.1038/nature19797

In this Letter, Supplementary Figs 10 and 12 were not updated. The analyses, calculations, and text are correct. The Supplementary Information to this Corrigendum contains the corrected Supplementary Figs 10 and 12.

In Supplementary Fig. 10a the blue line was about 30 Tg of CH₄ per year too low, and represents results from a previous model version. Supplementary Fig. 10b shows the percentage contributions from Supplementary Fig. 10a, so both panels contain the same error. The total of all lines in Supplementary Fig. 10a should sum to about 150 Tg of CH₄ per year, consistent with Fig. 2b. In Supplementary Fig. 12, the graph contains data from the literature (coloured points) as well as a distribution based on these data (grey and black lines) and the y-axis of the upper panel was accidentally shifted (the coloured points should have the same values as those in the cited literature). These small errors affect only Supplementary Figs 10 and 12 and do not propagate into other results. The correct data were used for the remainder of the model and the Letter.

We thank V. Petrenko and F. George for bringing this to our attention. The errors have not been corrected online.

Supplementary Information is available in the online version of the Corrigendum.